Principal Components of US Sector Prices and Macroeconomic Data

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Macroeconomic conditions and stock prices

- Chen et al. (1986) show that macroeconomic variables, among them industrial production, risk premium changes, yield curve twists and inflation, are priced in US stock returns.
- Macroeconomic indicators are numerous and strongly correlated with one another, as are the sectors that make up the stock market.
- The high dimensionality and multicollinearity of both sets obscure the contribution of any single variable.
- Principal component analysis (PCA) addresses this by an orthogonal transformation of correlated variables into uncorrelated principal components (PCs). The components are ordered by the share of variance each explains.

Research question

Question

1. How many components summarize 10 US macroeconomic indicators and 11 sector prices?
2. What do those components represent?
3. How much of the S&P 500 return do the sector components explain?

Finding

1. Two components summarize each dataset and explain 78.26% of the macroeconomic variance and 93.62% of the sector price variance.
2. The first macroeconomic component captures the joint movement of inflation and consumption, and the sector components are a market-wide factor and an energy factor.
3. The regression on the first two sector components explains 97.14% of the S&P 500 return variance in the training sample, with a lower testing error than on the first alone.

Macroeconomic data

Variable	Series
CPI	Consumer Price Index
Core CPI	Consumer Price Index less food and energy
PCE	Personal consumption expenditures
Retail Sales	Retail trade, excluding food services
Durable Goods Orders	New orders for durable goods
UNP	Unemployment rate
Wage Growth	Median hourly wage growth, nominal
PPI	Producer Price Index, all commodities
Industrial Production	Industrial production index
Trade Balance	Goods and services balance, USD million

Macroeconomic indicators.

- Monthly from January 2010 to September 2023.
- UNP and Trade Balance enter as levels.
- The others enter as year-on-year growth rates.

Sector and index data

- We use daily adjusted closing prices of the S&P 500 index and the 11 Select Sector SPDR exchange-traded funds that partition it.
- Training sample 19 June 2018 to 30 December 2022.
- Testing sample 3 January 2023 to 2 November 2023.

- Of the 55 sector price pairs, 85.5% have a correlation above 0.5, and energy (XLE) is the least correlated, from -0.08 with communication services (XLC) to 0.53 with financials (XLF).
- Of the 45 macroeconomic pairs, 55.6% are positive, and 60.0% of the positive pairs exceed 0.5.

Principal component analysis

$$R = \frac{1}{n-1} Z^\top Z = V \Lambda V^\top, \qquad \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_p), \quad \lambda_1 \geq \cdots \geq \lambda_p$$

$$y_{k,t} = \sum_j v_{jk} z_{tj}$$

- $X \in \mathbb{R}^{n \times p}$ holds n observations x_{tj} of p variables.
- Z holds the standardized variables $z_{tj} = (x_{tj} - \bar{x}_j)/s_j$, with $\bar{x}_j$ and s_j the mean and standard deviation of variable j.
- R is the correlation matrix and $V \Lambda V^\top$ its eigendecomposition.
- The columns $\mathbf{v}_k$ of V, with entries v_{jk}, are the loadings.
- The kth component $y_{k,t}$ has variance λ_k, and the components are mutually uncorrelated.

Selection of components

Three rules determine how many components to retain.

- Kaiser rule — Component k is retained when $\lambda_k > 1$, that is, when it explains more variance than a single standardized variable (Kaiser, 1960).
- Scree plot — The plot of λ_k against k, in which the components before the eigenvalues level off are retained (Cattell, 1966).
- Pareto plot — The cumulative share of variance, of which the retained components should capture at least 75%.

Principal component regression

$$r_t = \beta_0 + \sum_{k=1}^{m} \beta_k \, y_{k,t} + \varepsilon_t, \qquad m \in \{1, 2\}$$

$$\mathrm{RMSEP}(m) = \sqrt{\frac{1}{n} \sum_{t=1}^{n} \left(r_t - \hat{r}_t^{(-t)}(m) \right)^2}$$

- Model m, the principal component regression (PCR) of Massy (1965), regresses the S&P 500 log return r_t on the first m components by ordinary least squares.
- $y_{k,t}$ is the kth component of the 11 standardized sector log returns.
- Cross-validation over 10 random training segments (Mevik and Wehrens, 2007) assesses the number of components by the root mean squared error of prediction (RMSEP).
- $\hat{r}_t^{(-t)}(m)$ is fitted without the segment containing t.

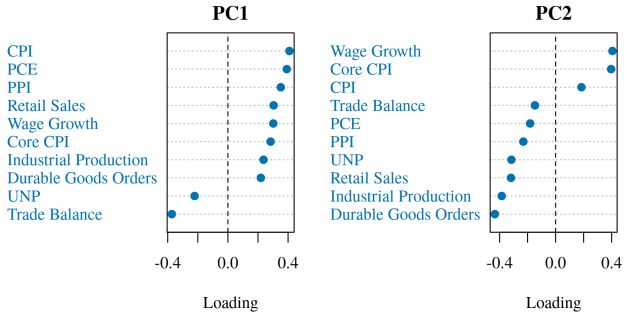
Leading eigenvalues

	Eigenvalue			Cumulative share (%)		
	PC1	PC2	PC3	PC1	PC2	PC3
Macroeconomic data	4.85	2.97	0.86	48.52	78.26	86.86
Sector prices	9.04	1.26	0.38	82.14	93.62	97.07

Leading eigenvalues and cumulative share of variance.

- In both datasets, $\lambda_1, \lambda_2 > 1 \geq \lambda_3$, and the eigenvalues level off after the second component (Appendix A).
- The first two components explain 78.26% of the macroeconomic variance and 93.62% of the sector price variance.
- Two components are therefore retained in each dataset.

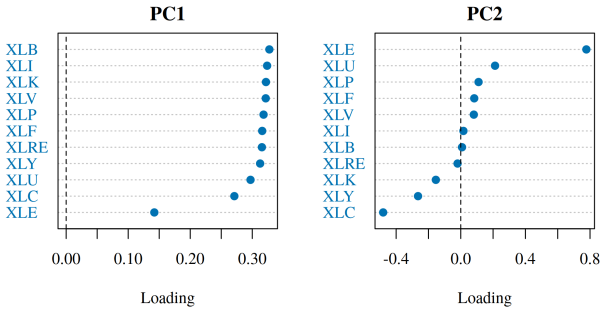
Loadings of the macroeconomic components



Loadings of the first (left) and second (right) macroeconomic components.

- PC1 loads positively on inflation and consumption, led by CPI at 0.41, PCE at 0.39 and PPI at 0.35. We attribute this to pandemic savings and a "You Only Live Once" attitude.
- PC2 contrasts Wage Growth at 0.41 and Core CPI at 0.40 with Durable Goods Orders at -0.43 and Industrial Production at -0.38.

Loadings of the sector components



Loadings of the first (left) and second (right) sector components.

- PC1 is a market-wide factor that aligns with overall market sentiment, with loadings between 0.27 and 0.33 on every sector except energy, which loads at 0.14.
- PC2 is an energy factor, with a loading of 0.78 on energy, while the loadings on the other sectors range from -0.48 for communication services to 0.21 for utilities (XLU).

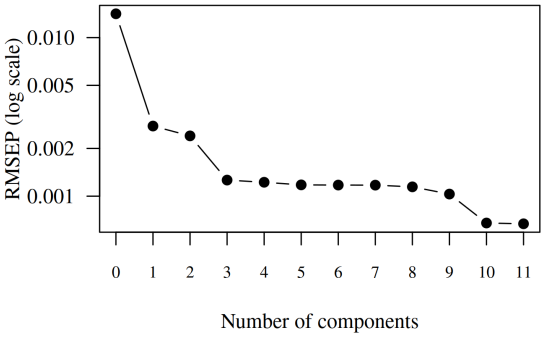
Variance preserved and explained

Components	1	2	3	4	5	6	7	8	9	10	11
Sector returns	72.50	80.60	87.69	90.36	92.65	94.54	95.96	97.26	98.46	99.31	100.00
S&P 500	96.19	97.14	99.21	99.26	99.31	99.32	99.32	99.36	99.49	99.78	99.78

Cumulative share (%) of the sector return variance preserved and of the S&P 500 return variance explained.

- The first component preserves 72.50% of the sector return variance and explains 96.19% of the index return variance.
- The second adds 8.10% of the sector variance but only 0.95% of the index variance.
- We therefore compare Model 1 with Model 2 to assess whether the second component improves the regression or leads to overfitting.

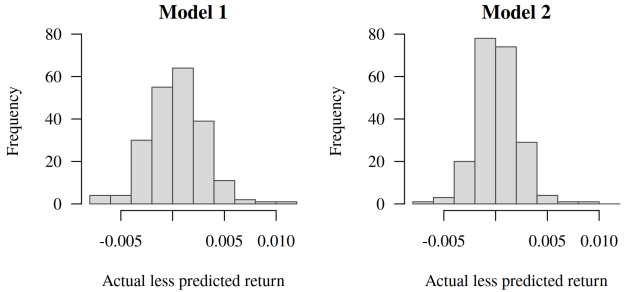
Cross-validated RMSEP



Cross-validated RMSEP against the number of components, on a logarithmic scale.

- The RMSEP falls 1.41% $\rightarrow$ 0.277% $\rightarrow$ 0.240% $\rightarrow$ 0.126% with 0, 1, 2 and 3 components.
- The first component delivers most of the reduction.

Testing errors



Distribution of the testing errors, defined as the actual less the predicted S&P 500 return.

- The root mean squared error is 0.206% under Model 2 against 0.266% under Model 1.
- Model 2 also concentrates its errors near zero, with 72.0% of testing days within 0.2% of the actual return against 56.4% under Model 1.
- Together with the training results, these findings indicate that Model 2 is the better model.

Conclusion

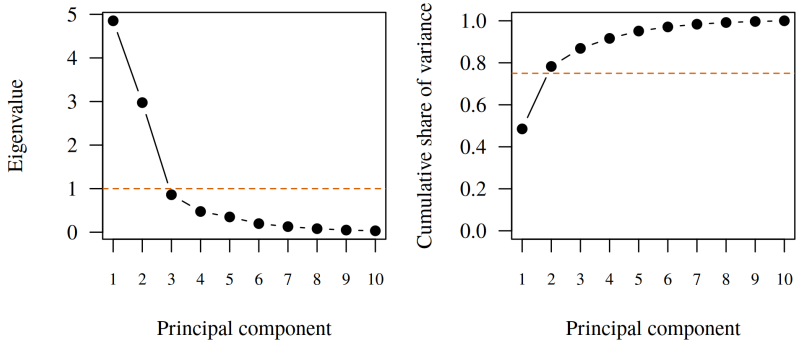
- We apply PCA to 10 US macroeconomic indicators and 11 sector prices, and two components summarize each dataset.
- The macroeconomic components describe inflation moving with consumption, and wages and core inflation set against durable goods orders and industrial production.
- The sector components describe a market-wide factor driven by sentiment and an energy factor driven by supply instability and robust demand expectations in the oil market.
- A PCR on the first two sector components explains 97.14% of the S&P 500 return variance in training and outperforms the regression on the first component in testing.

References

- Cattell, R. B. (1966). The scree test for the number of factors. *Multivariate Behavioral Research*, 1(2), 245–276.
- Chen, N.-F., Roll, R. and Ross, S. A. (1986). Economic forces and the stock market. *Journal of Business*, 59(3), 383–403.
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- Mevik, B.-H. and Wehrens, R. (2007). The pls package. Principal component and partial least squares regression in R. *Journal of Statistical Software*, 18(2), 1–23.

Code and replication data are available at *GitHub*.

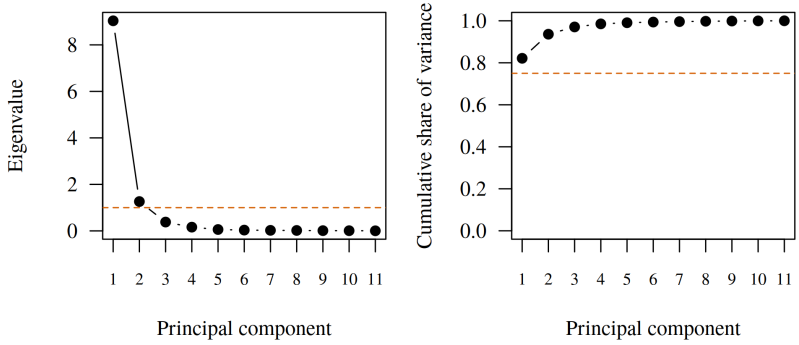
Appendix A.1. Macroeconomic scree and Pareto plots



Scree plot (left) and cumulative share of variance (right) of the macroeconomic data.

- The eigenvalues level off after the second component.
- The first two components explain 78.26% of the macroeconomic variance.

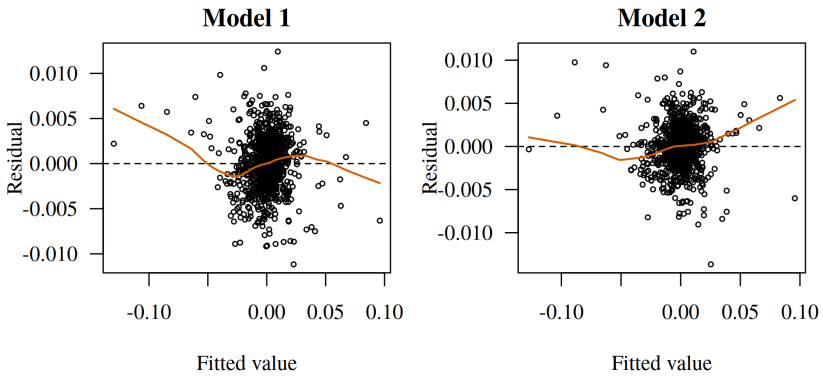
Appendix A.2. Sector price scree and Pareto plots



Scree plot (left) and cumulative share of variance (right) of the sector prices.

- The eigenvalues level off after the second component.
- The first two components explain 93.62% of the sector price variance.

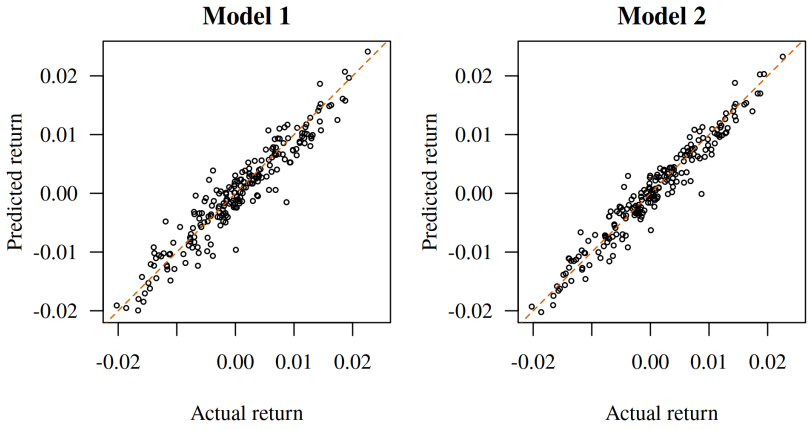
Appendix B.1. Residuals against fitted values



Residuals against fitted values. The solid line is a locally weighted smoother.

- The residual standard deviation is 0.239% under Model 2 versus 0.276% under Model 1.
- Model 2's fitted values therefore lie closer to the observed returns.

Appendix B.2. Predicted against actual returns



Predicted against actual S&P 500 log returns over the testing sample.

In the testing sample, both models track the index return closely.